Position, distance, displacement

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Learning target: Distinguish a location, a total path length, and a change in position.

Suggested first pass: 1, 2, 3, 9. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 1; 1 course days later.*
38 m if the 5 m/s rightward motion continues. Check a later timed position against the prediction.

02. Explain a worked example (worked)
Route arrows 4 m right and 2 m left; resultant 2 m right.

03. Read the route (model)
Start 1 m, turn 6 m, end 3 m.

04. Count the whole path (practice)
5 m + 3 m = 8 m.

05. Connect start to finish (reason)
Displacement 2 m right; final position 3 m. They differ because the start was 1 m, not zero.

06. A complete trip back (reason)
Distance 10 m; displacement 0 m because start and end coincide.

07. Fix the wording (reason)
No. Distance is path length. Example: 4 m out and 4 m back gives 8 m distance and zero displacement.

08. Choose a method (mixed) *Review: Unit 1, Day 1; 1 course days later.*
Position increases by 5 m each second; equal rightward changes suggest steady motion.

09. Exit check (exit)
Distance 5+2=7 m; displacement 3 m right; final position 5 m.

10. STRETCH: Apply it another way (stretch)
Direct 1 to 4: 3 m. Route 1 to 6 to 4: 5+2=7 m. Both displacements 3 m right.

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Suggested first pass: 1, 2, 3, 9. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 1; 1 course days later.*
v=-2.5 m/s; x(7)=-2.5 m. Assumes constant velocity; compare another timed position with its uncertainty.

02. Explain a worked example (worked)
Route arrows 4 m right and 2 m left; resultant 2 m right.

03. Track a signed route (model)
Distance 7+5=12 m; displacement -1-(-3)=+2 m.

04. Change the coordinate convention (practice)
Coordinates become +3,-4,+1 m; displacement -2 m. Physical route, distance 12 m, and displacement direction are unchanged.

05. A closed path (reason)
No. Distance is 500 m. Need geometry/start/end positions; displacement is the chord from start to end, not the arc length. Full lap contributes zero displacement.

06. Different paths, same endpoints (reason)
Examples: east then north gives 7 m; direct diagonal gives 5 m. Both displacement (4, 3) m, magnitude 5 m.

07. Zero and nonzero (reason)
Yes: return trip. No: nonzero displacement requires a path of nonzero length; distance is at least displacement magnitude.

08. Choose a method (mixed) *Review: Unit 1, Day 1; 1 course days later.*
Motion is reported across 3 images. Uncalibrated images do not establish size, distance, full trajectory or impact; more measurements and uncertainties are needed.

09. Exit check (exit)
Distance 7+3=10 m; displacement 1-5=-4 m; final position +1 m. They describe path length, change, and location respectively.

10. STRETCH: Apply it another way (stretch)
Turn at +6 m: 8+3=11 m; or -5 m: 3+8=11 m. Displacement +5 m in either case.